LESSON 14 · MISSION GUIDE · THE FINALE

# The Charter

Two sessions. One report. Everything you've learned, in one project folder.

From **Queen Aldara of Thornmere**
To **Jen, Keeper of the Guild Ledger**
By raven **Midsummer approaches. The good biscuits.**

Keeper. Fourteen lessons ago we told you we did not know what a Python was. We still do not, strictly, but we have seen the charts, Orrin says the Vault answers you now, and we are no longer afraid of the full moon.

Here is our command. At Midsummer we renew the guild's Royal Charter, or we do not. Before we sign, the Privy Council shall have, properly done: how often the guild wins, which quests are worth the crowns, which are too dangerous, and whether the loot is shared out fairly. A picture, a workbook the councillors can take home in their carriages, and **one sentence we can say out loud**. The Grimoire has told the Council for years that the guild wins **85%** of its quests and should hunt nothing but wyrms. No one has ever been able to explain why.

If your numbers hold, the Charter is renewed and the Grimoire is closed, for good. Quartermistress Pell has baked a small cake. Biscuit has been told not to light it. No pressure. (Some pressure.) — A.

## What you'll build

```
lessons/14-the-charter/
├── data/guild.db + quests.csv, heroes.csv, party.csv, regions.csv
└── my_work/
    ├── charter.py              your functions: SQL and pandas, never print
    ├── check1.py … check5.py   one tiny script per exercise: import charter, print
    ├── report_console.py       session 2: the report on screen
    ├── report.py               session 2: the report, a PNG and a workbook
    └── output/                 crowns_per_day.png, charter_report.xlsx
```

## What you'll be able to do after today

- [ ] Answer a real question end to end, from a **database and CSV files** to a chart, a workbook and a sentence
- [ ] Choose the right craft: **SQL** (`read_sql`, JOIN, LEFT JOIN) where the Vault is good at it, **pandas** where it's easier
- [ ] Structure a real piece of work as a **module of functions** plus small scripts that use it
- [ ] Pick the measure that tells the truth (**paid** per day, not the reward on the notice) and say why
- [ ] Ship it: a chart, a formatted workbook, a pull request with a title that means something

### Why this matters at work

- **This is the actual job.** Every analytics role is this loop: data in a database and some files, a question from someone important, a number that holds up, a chart, a workbook, one sentence. You'll have done it end to end in a couple of hundred lines.
- **Functions are how you stay honest.** One `success_rate` used everywhere means one place to be right. The Grimoire had the incantation copied out eleven times, three of them differently.
- **Next season, it's one command.** New quests in the Vault, `python my_work/report.py`, done.

## How the two sessions work

SESSION 1
**The numbers**
15 min briefing, then exercises 1–4: outcomes, difficulty, crowns per day, regions

SESSION 2
**The report**
10 min recap, then exercises 5–7: loot and fairness, the console report, chart + workbook

LAST 15 MIN
**The Grimoire's last page**
Its own Charter figures. Find out why it said 85%.

### Things that will bite you

- **Fled is not a success.** It pays a quarter, and it's still running away.
- A reward on the notice board is not crowns in the chest. Measure `paid`, and per **day**, or the Wyrm wins.
- A plain `JOIN` drops rows without a partner. Thornwood has no warden; `LEFT JOIN` keeps it.
- `charter.py` never prints. If you want to see something, print it in a check script.

### You're done when...

- [ ] `python my_work/report.py` prints the report on the sheet and writes the PNG and the workbook
- [ ] `charter.py` uses at least two SQL queries, every function has a docstring, and no function prints
- [ ] The Grimoire's last page agrees with you, and your three commit messages say exactly why it didn't before
- [ ] The pull request is called **The Charter** and contains the Queen's sentence

---




**After this:** there is no lesson 15. There's Midsummer, a small cake, a Charter with a royal seal on it, and a very old book being closed. The Queen will say one sentence to the Privy Council: *"The guild wins 65% of its quests, and Trolls pay best at 76 crowns a day; the Wyrm, for all its 3,130-crown rewards, earns just 29."* Then there's every other database and spreadsheet in the realm, and you know what to do with them now.